

# Topic 5: Segmentation

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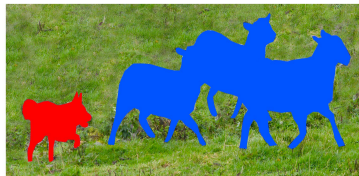
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Chennai-127

- 1 What is segmentation
- 2 A simple algorithm to detect characters in number plate
- 3 Network Flow
  - Ford-Fulkerson algorithm to find Max Flow
  - Max-Flow Min-Cut Theorem
  - Image Reconstruction using Max-Flow
  - Graph Cut Algorithm: Image Segmentation using Min-Cut
- 4 Mean shift segmentation algorithm
- 5 Segmentation using K-Means Clustering

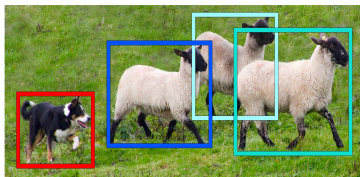
- Segmentation aims to create meaningful regions in an image, ideally aligned with objects



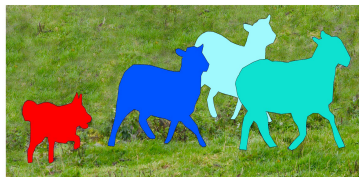
Image Recognition



Semantic Segmentation



Object Detection

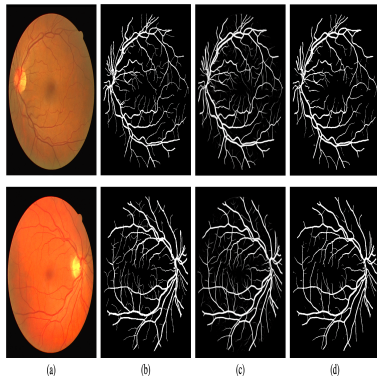
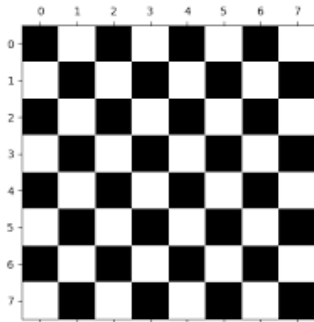


Instance Segmentation

- ▶ Let  $I$  be the image,  $R$  be the set of pixel locations in  $I$
- ▶ The segmentation of image  $I$  is the partition of  $R$  into  $(R_1, R_2, \dots, R_k)$  with the following conditions
  - $R = \cup_{i=1}^k R_i$
  - $R_i \cap R_j = \phi$  for all  $i \neq j$
  - For each  $i$ ,  $R_i$  is connected
  - For each  $i$ ,  $Q(R_i)$  is true, where  $Q$  is some predicate on similarity
  - For each  $i, j$ , If  $Q(R_i) = \text{True}$ , and  $Q(R_j)$  is true, and also  $R_i$  and  $R_j$  are adjacent, then  $Q(R_i \cup R_j)$  is false

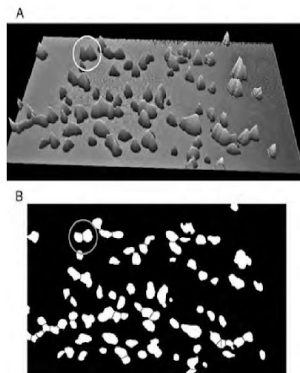
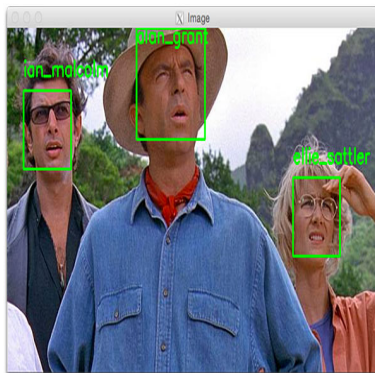


# Examples of segmentation



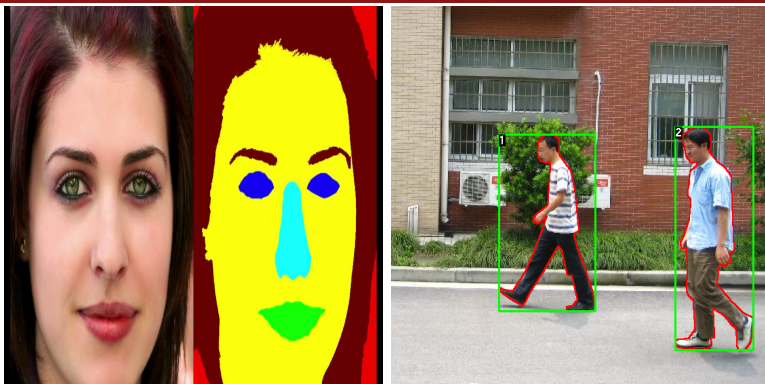
- ▶ In Chessboard image, each square( black/white) is a segment
- ▶ For the retina, white pixels constitute a region, black pixels constitute another region

# Examples of segmentation (cont.)



- ▶ Face detection involves segmentation( Detection: Segment and classify)
- ▶ Face recognition is a classification problem

# Examples of segmentation (cont.)



## The semantics of segments are problem dependent

- ▶ In the same input image, whole face can be a single segment or each part may be a single segment
- ▶ In the same input image, whole person may a single segment or face alone may be a single segment

## Algorithm to segment the characters in the number plate

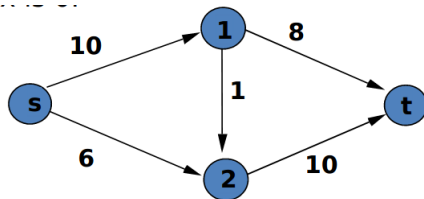
- ▶ Convert RGB to gray scale image
- ▶ Binarize the gray scale image using thresholding
- ▶ Find connected component(CC), and declare each CC as a segment



## Definition of flow network:

Flow network is a directed weighted graph  $G=(V,E,c)$  such that

- ▶ Weight(capacity)  $c(u, v) \geq 0$ .
- ▶ Two distinguished vertices exist in  $G$  namely :
  - Source (denoted by  $s$ ) : In-degree of this vertex is 0.
  - Sink (denoted by  $t$ ) : Out-degree of this vertex is 0.





## Definition of flow

Flow in a network is an integer-valued function  $f$  defined on the edges of  $G$  satisfying:

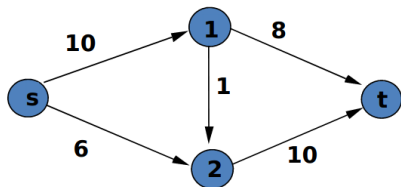
- ▶  $0 \leq f(u, v)$ , for every edge  $(u, v)$  in  $E$ .
- ▶ Capacity constraint:  $\forall u, v \in V, f(u, v) \leq c(u, v)$
- ▶ Skew symmetry:  $\forall u, v \in V, f(u, v) = -f(v, u)$
- ▶ Flow conservation: For each vertex  $v$  other than  $s$  and  $t$ ,  $\text{inflow}(v) = \text{outflow}(v)$

Skew symmetry condition implies that  $f(u, u) = 0$

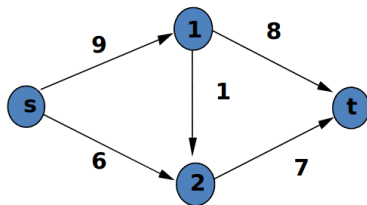
# Network Flow (cont.)



Flow Network:



Flow:





## Definition of Max Flow

- ▶ The value of a flow is given by :

$$|f| = \sum_{v \in V} f(s, v) = \sum_{v \in V} f(v, t)$$

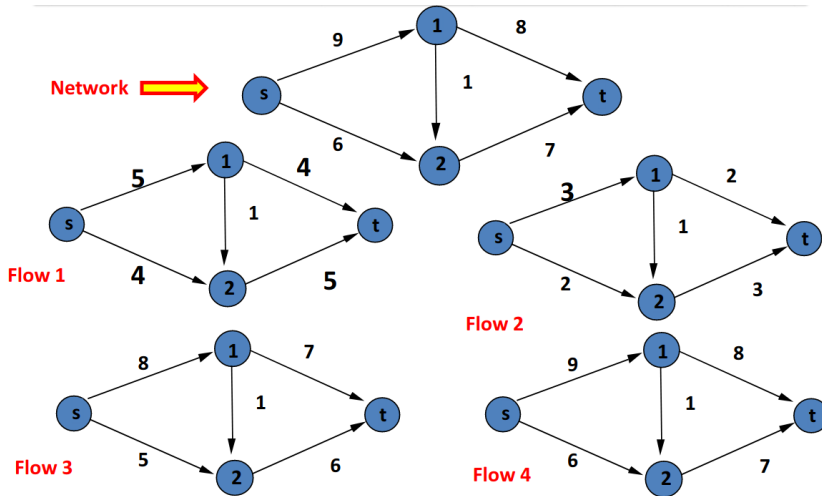
- ▶ Definition of max flow:
  - Given a graph  $G=(V, E)$  with capacities on edges, find flow  $f$ , such that  $|f|$  is maximum.



# Network Flow (cont.)



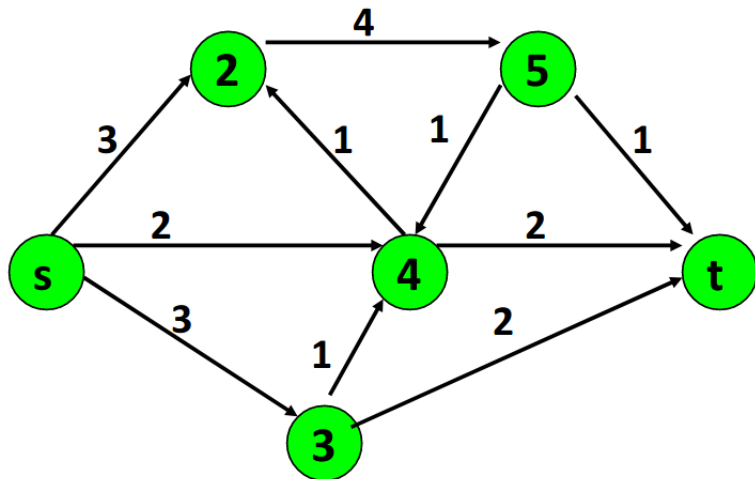
## Example of a Max Flow



# Ford-Fulkerson algorithm to find Max Flow



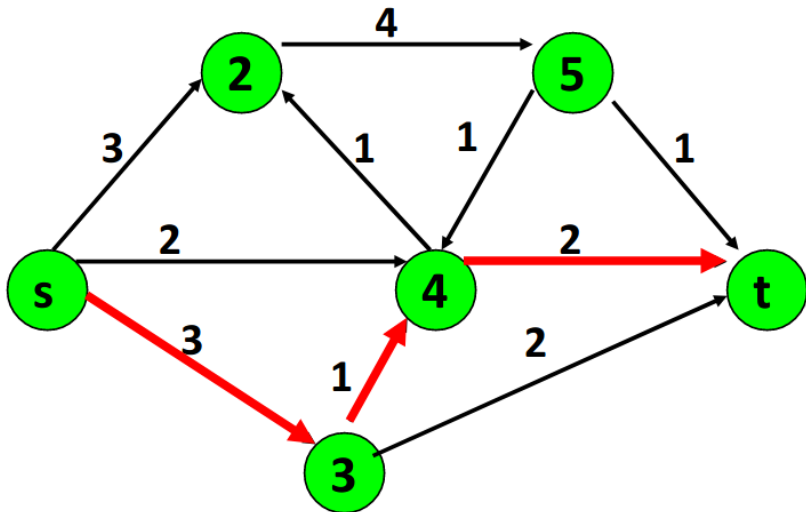
Flow network



# Ford-Fulkerson algorithm to find Max Flow (cont.)



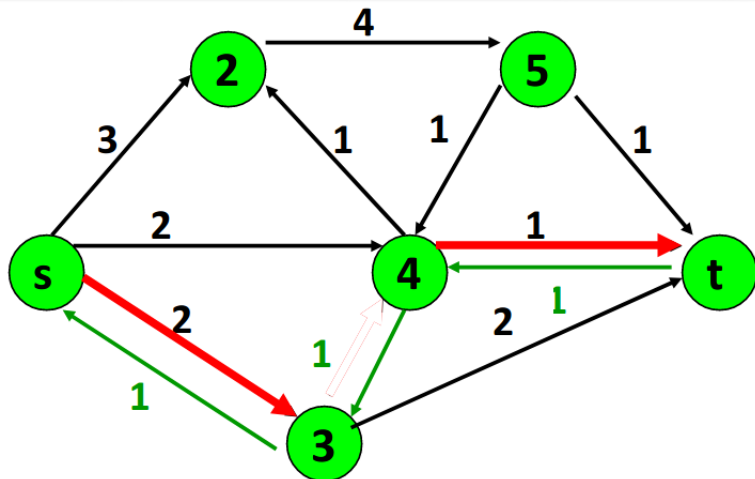
Find any s-t path in  $G(x)$



# Ford-Fulkerson algorithm to find Max Flow (cont.)



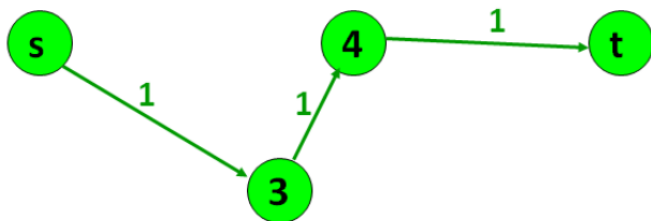
Residual graph:



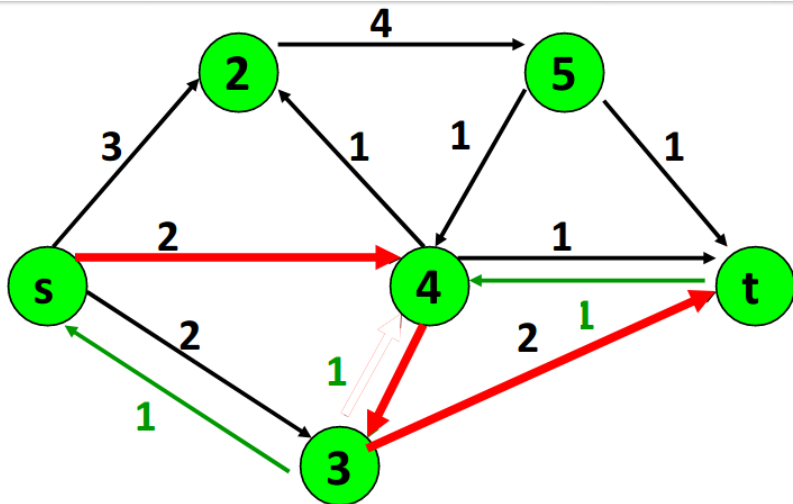
# Ford-Fulkerson algorithm to find Max Flow (cont.)



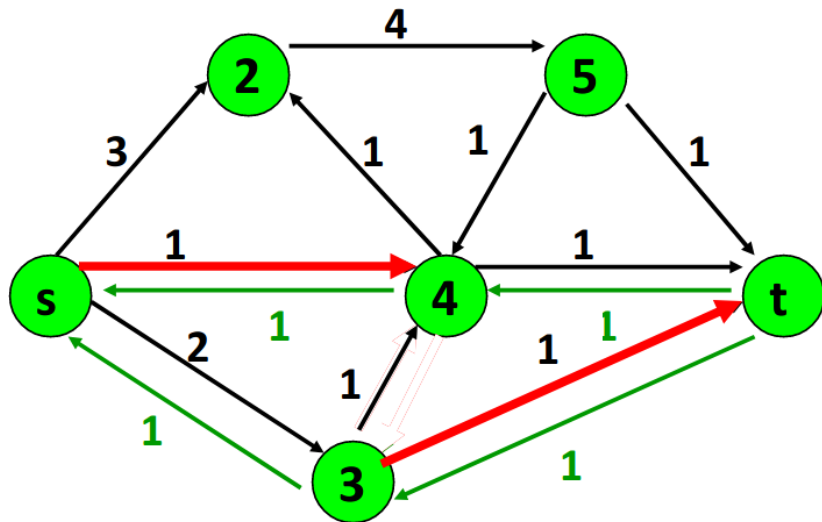
Flow:



# Ford-Fulkerson algorithm to find Max Flow (cont.)



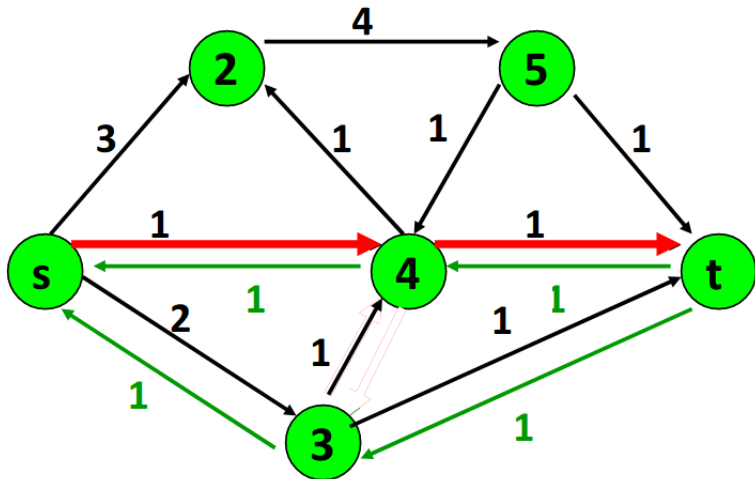
# Ford-Fulkerson algorithm to find Max Flow (cont.)



# Ford-Fulkerson algorithm to find Max Flow (cont.)



Find any s-t path

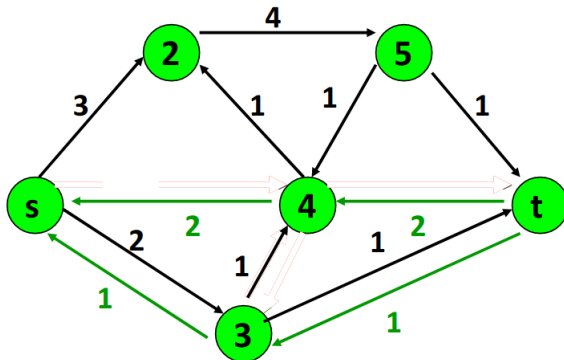




# Ford-Fulkerson algorithm to find Max Flow (cont.)



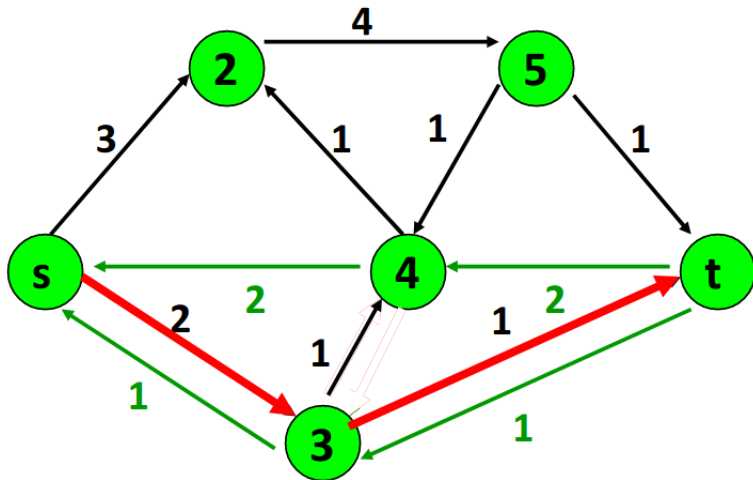
- Determine the min capacity  $\Delta$  of the path
- Send  $\Delta$  units of flow in the path
- Update residual capacities



# Ford-Fulkerson algorithm to find Max Flow (cont.)



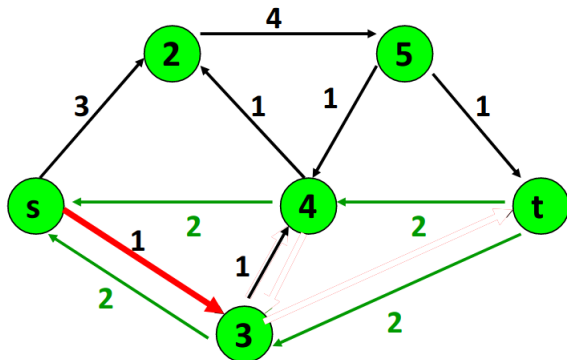
Find any s-t path



# Ford-Fulkerson algorithm to find Max Flow (cont.)



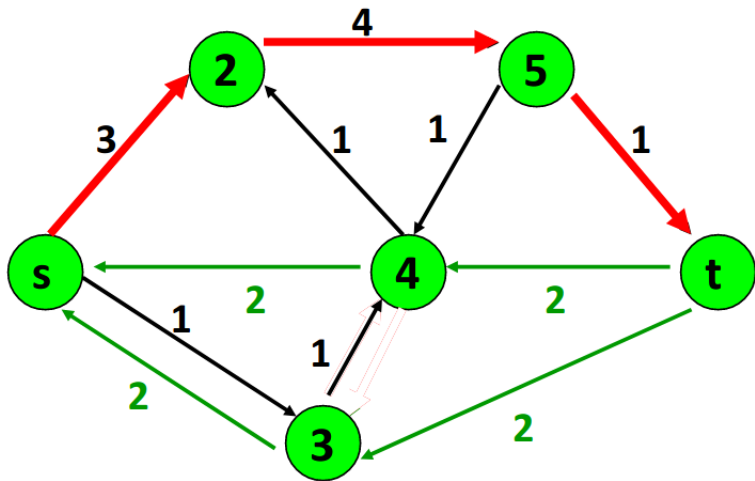
- Determine the capacity  $\Delta$  of the path
- Send  $\Delta$  units of flow in the path
- Update residual capacities



# Ford-Fulkerson algorithm to find Max Flow (cont.)



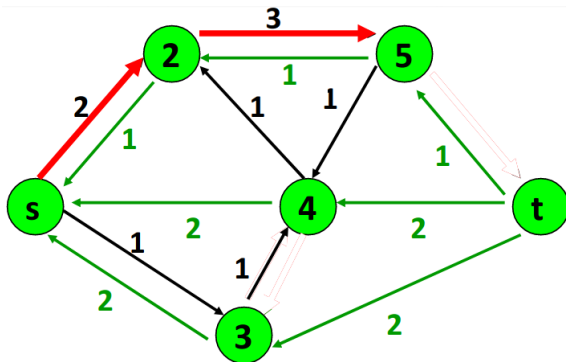
Find any s-t path



# Ford-Fulkerson algorithm to find Max Flow (cont.)



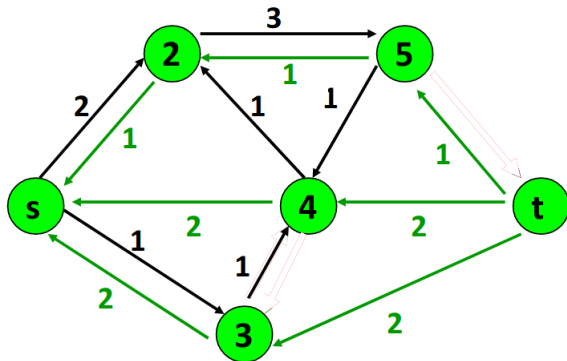
- Determine the capacity  $\Delta$  of the path
- Send  $\Delta$  units of flow in the path
- Update residual capacities



# Ford-Fulkerson algorithm to find Max Flow (cont.)



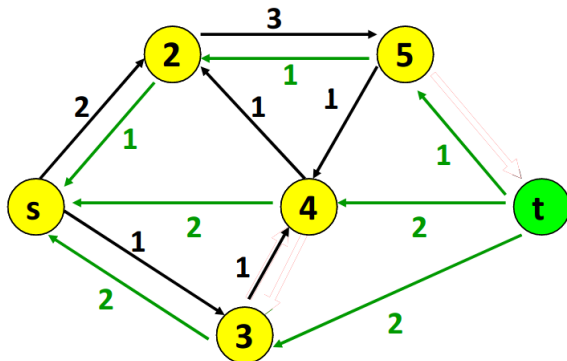
- There is no s-t path in the residual network.



# Ford-Fulkerson algorithm to find Max Flow (cont.)



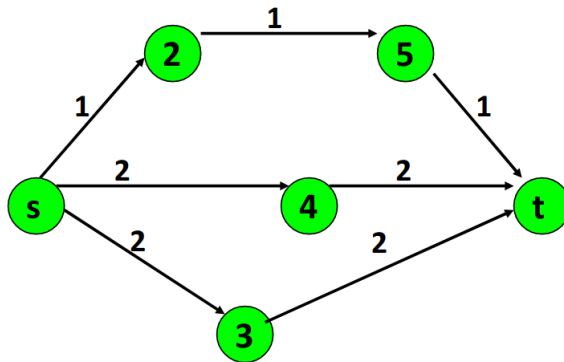
- These are the nodes that are reachable from node s.



# Ford-Fulkerson algorithm to find Max Flow (cont.)



- Here is the optimal flow





# Steps of Ford Fulkerson Algorithm



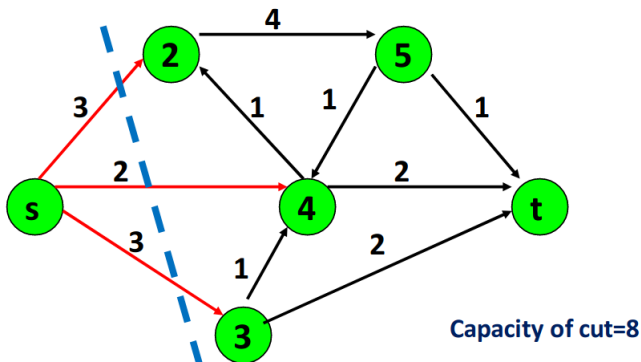
**Input  $G=(V,E,C),s,t$**

Assume  $G = (V, V \times V, C)$ , where  $C(u,v)=0$  if  $(u, v)$  is not in  $E$

- ▶ Find s-t path in  $G$
- ▶ Find min weight in the s-t path, say  $a$
- ▶ Save the flow with the s-t path with weight  $a$
- ▶ Find residual graph  $G' = (V, V \times V, C')$   
where
  - $C'(u, v) = C(u, v) - a$  when  $(u,v)$  is in the path
  - $C'(v, u) = C(v, u) + a$  when  $(u,v)$  is in the path
  - $C'(u, v) = C(u, v)$  when  $(u,v)$  is not in the path
- ▶  $G = G'$
- ▶ Repeat the above five steps until there is no s-t path
- ▶ output the accumulated flow

## Definition of s-t Cut

- Set of edges is said to be a cut if they are removed from the graph, graph should be disconnected into two components, one with source and the other one with sink.

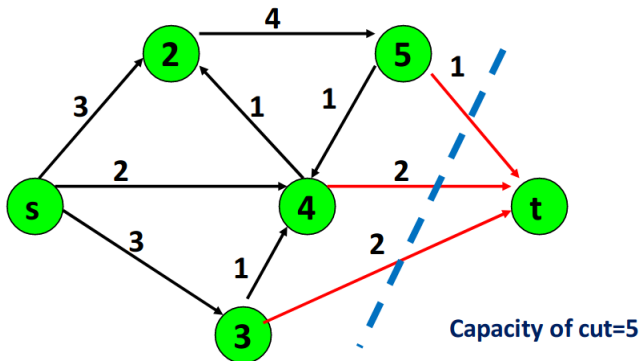


## Max-Flow and Min-cut Theorem

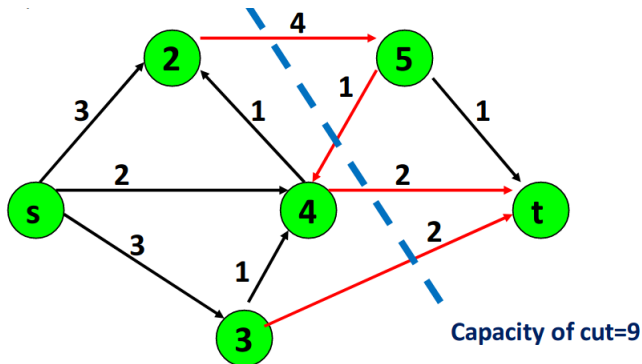


- ▶ The maximum flow value is the same as the minimum cut value

# Max-Flow Min-Cut Theorem (cont.)



# Max-Flow Min-Cut Theorem (cont.)



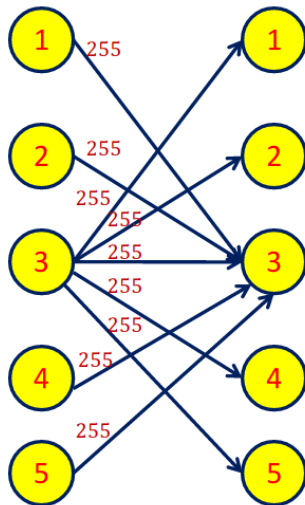
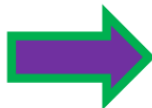
# Image Reconstruction using Max-Flow



# Image Reconstruction using Max-Flow (cont.)



|   | 1   | 2   | 3   | 4   | 5   |
|---|-----|-----|-----|-----|-----|
| 1 | 0   | 0   | 255 | 0   | 0   |
| 2 | 0   | 0   | 255 | 0   | 0   |
| 3 | 255 | 255 | 255 | 255 | 255 |
| 4 | 0   | 0   | 255 | 0   | 0   |
| 5 | 0   | 0   | 255 | 0   | 0   |



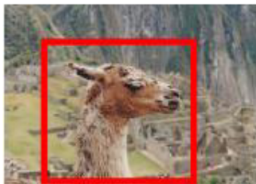
**Image**

**Graph representation**

# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut



## Image Segmentation

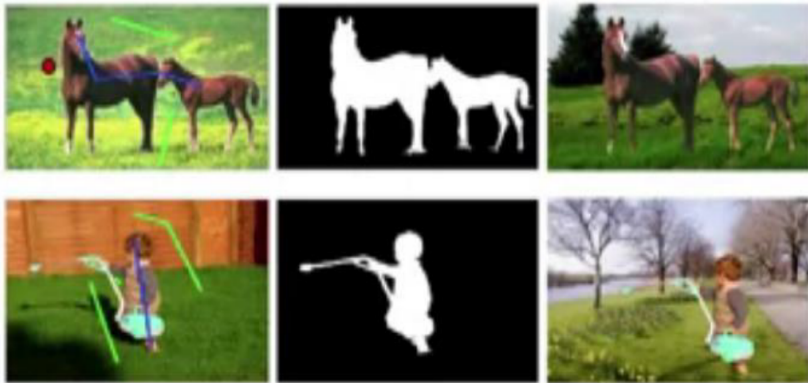




# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



## Image Segmentation

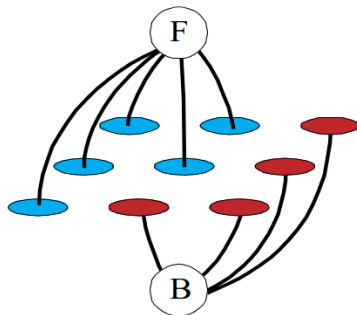


# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



- ▶ Each pixel = node
- ▶ Add two nodes F & B
- ▶ Labeling: link each pixel to either F or B

|   |   |   |
|---|---|---|
| F | F | B |
| F | F | B |
| F | B | B |

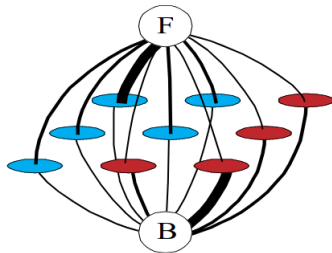


Desired result

# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



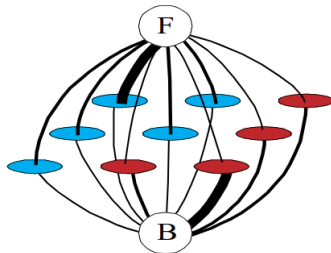
- Construct graph with data term
  - Put one edge between each pixel and F
  - Put one edge between each pixel and B
  - Weight of edge between  $i$  and F:  $w_{iF} = -\lambda \log(P_B(i))$
  - Weight of edge between  $i$  and B:  $w_{iB} = -\lambda \log(P_F(i))$



# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



- Add Smoothness term to the Graph
  - Add an edge between each neighbor pair (i,j)
  - Weight of edge between i and j:  $w_{ij} = \exp(-(I_i - I_j)^2 / 2\sigma^2)$

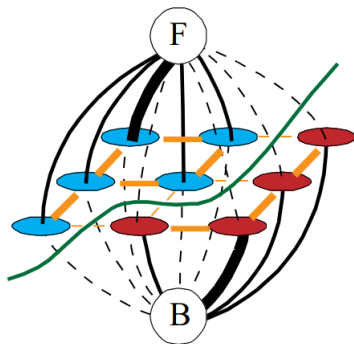


# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



## ► Min-Cut

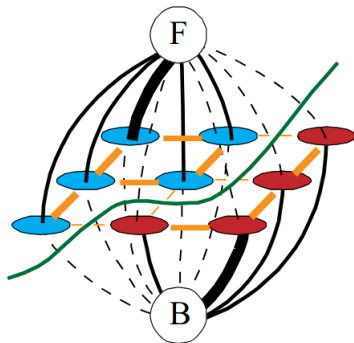
- Cut: Remove edges to disconnect F from B
- Min Cut: Cut with sum of its edge weights is minimum



# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



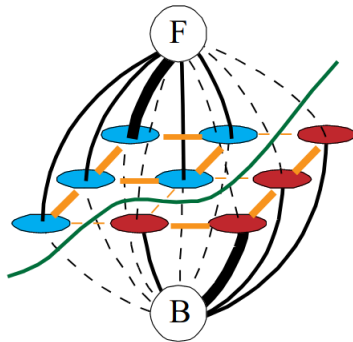
- ▶ In order to be a cut:
  - For each pixel, either the F or G edge has to be cut
- ▶ In order to be minimal
  - Only one edge to F or B per pixel can be cut



# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



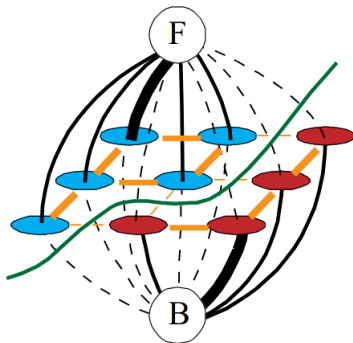
- Which edges are to be removed?
  - Edges with least cost



# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



- ▶ Since  $w_{iF} = -\lambda \log(P_B(i))$ , larger value of  $P_B(i)$  will lead to smaller value of  $w_{iF}$ 
  - Hence edge (i,F) will be removed, and edge (i,B) will stay
  - ie. Edge (i,B) will stay after cut if prob of i to be background is high

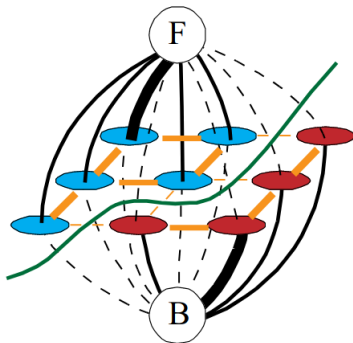




# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



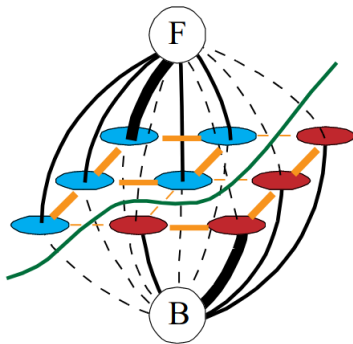
- ▶ Since  $w_{iB} = -\lambda \log(P_F(i))$ , larger value of  $P_F(i)$  will lead to smaller value of  $w_{iB}$ 
  - Hence edge (i,B) will be removed, and edge (i,F) will stay
  - ie. Edge (i,F) will stay after cut if prob of i to be background is high



# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



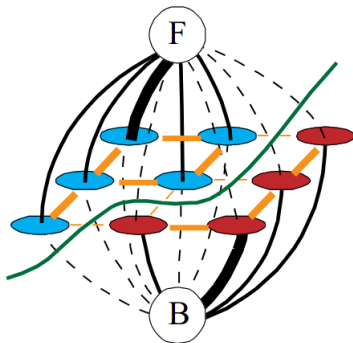
- ▶ Since for neighboring pixels (i,j),  $w_{ij} = \exp(-(I_i - I_j)^2 / 2\sigma^2)$ , similar i and j,  $w_{ij}$  is very high
- ▶ Hence edge between similar pixels will stay after graph cut.
- ▶ This provides smoothness



# Graph Cut Segmentation Algorithm:: Image Segmentation using Min-Cut (cont.)



- How to find min-cut
  - Apply max-flow algorithm
  - The output of max-flow algorithm will result in min-cut (max-flow min-cut theorem)

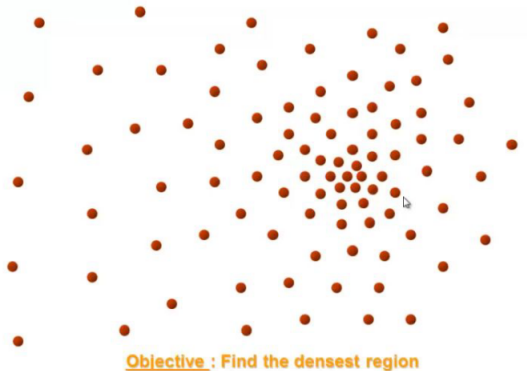




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**Mean Shift:** Given a distribution of points, and also a point  $x$ , mean shift is a procedure for finding the centroid of local densest region for  $x$ .





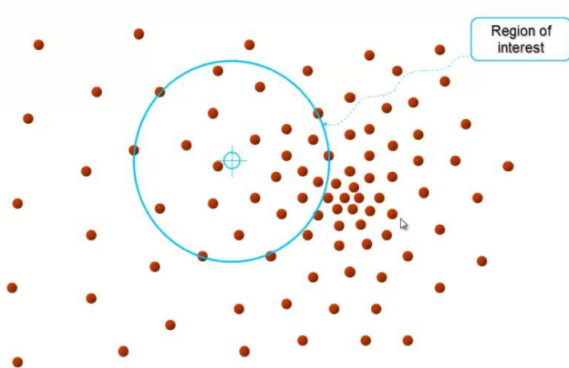
- ▶ The concept of mean shift as a **mode-seeking algorithm** proposed in 1975
  - Fukunaga, K., Hostetler, L. (1975). The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition. IEEE Transactions on Information Theory.
- ▶ Mean Shift segmentation algorithm was proposed in 2002.
  - "Mean Shift: A Robust Approach Toward Feature Space Analysis" Published in: IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), 2002.

# Mean shift segmentation algorithm (cont.)



**How to find centroid of local densest region for given  $x$**

**Step 1: Find the centroid of points centered at  $x$  with radius  $r$ , say such centroid as  $c$**  (Centroid of set of points is mean of those points)



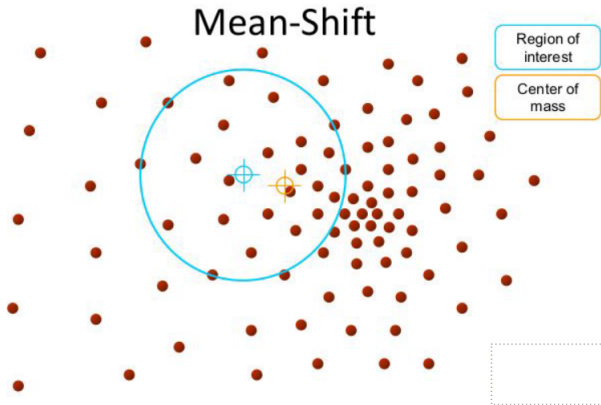
**Objective : Find the densest region**



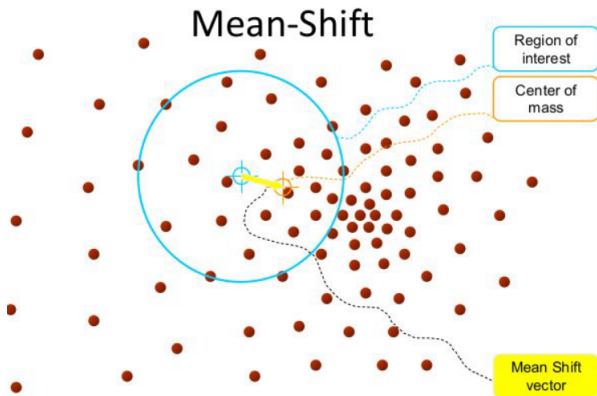
# Mean shift segmentation algorithm (cont.)



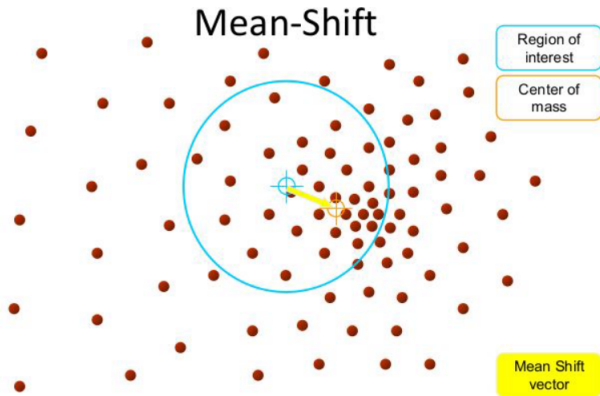
**Step 2: Find the centroid of points centered at  $c$  with radius  $r$ , say such centroid as  $c_{\text{prime}}$**



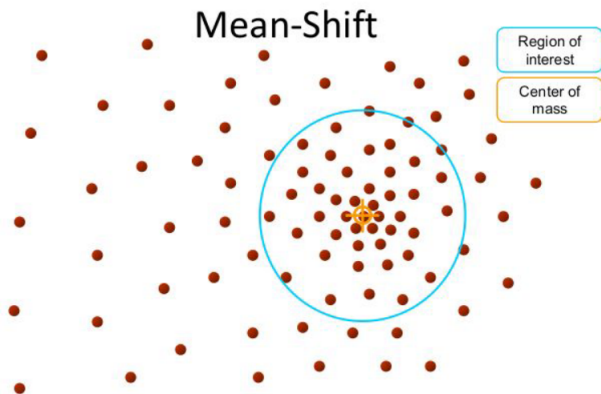
**Step 3:** If  $|c - c'| > T$ ,  $c = c'$  goto step 2



# Mean shift segmentation algorithm (cont.)



# Mean shift segmentation algorithm (cont.)





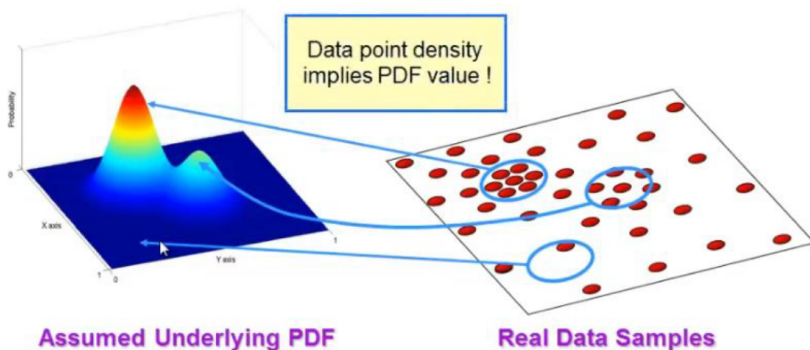
## Mean shift Segmentation Algorithm(Basic version)

**Input: Image; Output: Labeling of each pixel in the image**

- ▶ Find the feature vector for each pixel in the input image(eg.  $(r,g,b)$  is feature vector for a pixel at  $(x,y)$ )
- ▶ Considering each feature vector as a point in feature space, do the segmentation of points in the feature space using the following procedure
- ▶ Each point in the feature space
  1. Find the centroid of points centered at  $x$  with radius  $r$ , say such centroid as  $c$
  2. Find the centroid of points centered at  $c$  with radius  $r$ , say such centroid as  $c'$
  3. If  $|c - c'| > T$ ,  $c = c'$ ; goto step 2
  4. Assign a label  $l$  to the centroid  $c'$

5. Assign the label  $l$  to  $x$  also

- If a feature vector  $x$  is labeled with  $l$ , and  $x$  is the feature vector for pixel  $p$ , assign label  $l$  to  $p$





## Is weighted centroid better than centroid?

- ▶ Weighted centroid is the weighted mean
- ▶ Yes... weighted centroid is better
- ▶ what are the optimum weights?
- ▶ The preferred weights are expected to be inversely proportionate to the distance between the weighted centroid and the points in the set
  - Let us define  $f(y)$  as given below, where  $y$  which is maximizing  $f(x)$  is called as optimal weighted centroid

## How to find optimal weighted centroid

- ▶ Let

$$u_i(y) = \left\| \frac{x_i - y}{h} \right\|^2$$

# Mean shift segmentation algorithm (cont.)



- ▶ Gaussian kernel profile:

$$K(u) = C \exp\left(-\frac{u}{2}\right)$$

- ▶ Therefore,

$$K(u_i) = C \exp\left(-\frac{u_i}{2}\right)$$

and

$$K'(u_i) = -\frac{C}{2} \exp\left(-\frac{u_i}{2}\right) = -\frac{1}{2}K(u_i)$$

- ▶ Kernel density estimate:

$$f(y) = C \sum_{i=1}^N K(u_i(y))$$

- ▶ To find a maximum of  $f(y)$ :

$$\nabla f(y) = 0$$



# Mean shift segmentation algorithm (cont.)



$$\nabla f(y) = C \sum_{i=1}^N K'(u_i) \nabla u_i$$

$$u_i = \frac{\|x_i - y\|^2}{h^2}$$

$$\nabla u_i = \frac{2(y - x_i)}{h^2}$$

Hence,

$$\nabla f(y) = \frac{2C}{h^2} \sum_{i=1}^N (y - x_i) K'(u_i) = 0$$

Therefore,

$$\sum_{i=1}^N (y - x_i) K'(u_i) = 0$$

$$y \sum_{i=1}^N K'(u_i) = \sum_{i=1}^N x_i K'(u_i)$$

and hence,

$$y = \frac{\sum_{i=1}^N x_i K'(u_i)}{\sum_{i=1}^N K'(u_i)}$$

# Mean shift segmentation algorithm (cont.)



Thus,  $y$  is the weighted centroid, with weights determined by  $K'(u_i)$ .

For the Gaussian kernel, define positive weights as

$$w_i = -K'(u_i) = \frac{C}{2} \exp\left(-\frac{u_i}{2}\right)$$

so that

$$y = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i}$$

The equation for  $y$  is implicit because  $u_i$  depends on  $y$ :



$$u_i(y) = \left\| \frac{x_i - y}{h} \right\|^2$$

Hence, the stationary point  $y^*$  satisfies the fixed-point equation

$$y^* = \frac{\sum_{i=1}^N x_i K'(u_i(y^*))}{\sum_{i=1}^N K'(u_i(y^*))}$$

To obtain  $y^*$  iteratively, start with an initial estimate  $y^{(0)}$ .

At iteration  $j$ ,

# Mean shift segmentation algorithm (cont.)



$$u_i^{(j)} = \left\| \frac{x_i - y^{(j)}}{h} \right\|^2$$

$$w_i^{(j)} = -K'(u_i^{(j)})$$

and compute the new weighted centroid:

$$y^{(j+1)} = \frac{\sum_{i=1}^N w_i^{(j)} x_i}{\sum_{i=1}^N w_i^{(j)}}$$

Repeat until



$$\left\| y^{(j+1)} - y^{(j)} \right\| < \epsilon$$

At convergence,

$$y^{(j+1)} \approx y^{(j)} \approx y^*$$

and the mean-shift vector is

$$m(y^{(j)}) = y^{(j+1)} - y^{(j)}$$

**How to find mean shift with optimal weights?**

**Mean shift Segmentation Algorithm(Improved version)**

**Input: Image; Output: Labeling of each pixel in the image**

- ▶ Find the feature vector for each pixel in the input image(eg.  $(r,g,b)$  is feature vector for a pixel at  $(x,y)$ )
- ▶ Considering each feature vector as a point in feature space, do the segmentation of points in the feature space using the following procedure
- ▶ Each point in the feature space
  1. Using the following formula, find the weighted centroid of points centered at  $x$  with radius  $r$ , say such centroid as  $c$

$$c = \frac{\sum_i x_i K'(\|\frac{x_i - x}{h}\|^2)}{\sum_i K'(\|\frac{x_i - x}{h}\|^2)}$$



2. Using the following formula, find the centroid of points centered at  $c$  with radius  $r$ , say such centroid as  $c'$



$$c' = \frac{\sum_i x_i K'(\|\frac{x_i - c}{h}\|^2)}{\sum_i K'(\|\frac{x_i - c}{h}\|^2)}$$

3. If  $|c - c'| > T$ ,  $c = c'$ ; goto step 2
  4. Assign a label  $l$  to the centroid  $c'$
  5. Assign the label  $l$  to  $x$  also
- If a feature vector  $x$  is labeled with  $l$ , and  $x$  is the feature vector for pixel  $p$ , assign label  $l$  to  $p$



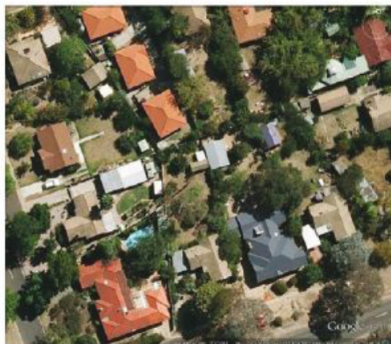
# Mean shift segmentation algorithm (cont.)



## Examples



## Examples



Original Satellite Imagery



Mean Shift Clustered Imagery

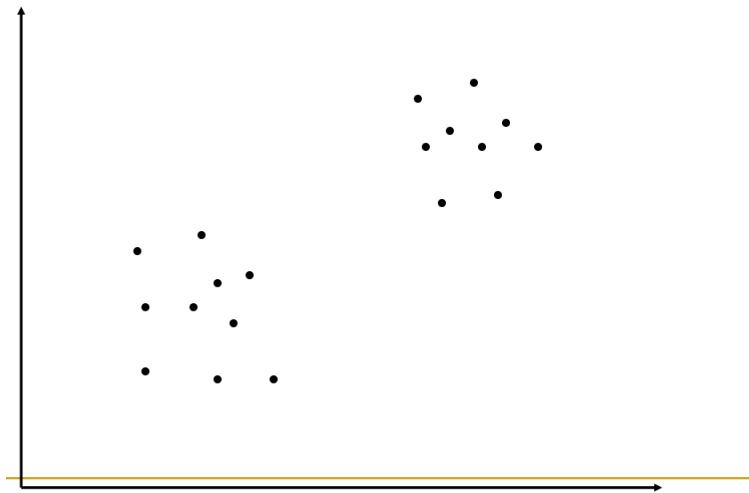
## Examples



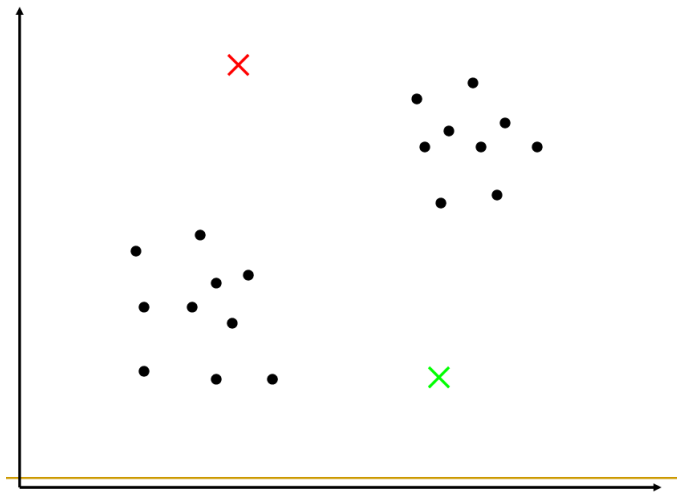


1. Partition the data points into  $K$  clusters randomly. Find the centroids of each cluster.
2. For each data point:
  - 2.1 Calculate the distance from the data point to each cluster.
  - 2.2 Assign the data point to the closest cluster.
3. Recompute the centroid of each cluster.
4. Repeat steps 2 and 3 until there is no further change in the assignment of data points (or in the centroids).

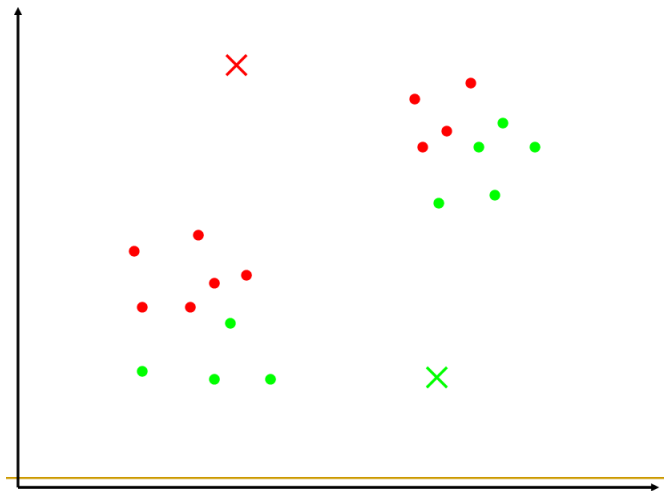
# Segmentation using K-Means Clustering (cont.)



# Segmentation using K-Means Clustering (cont.)

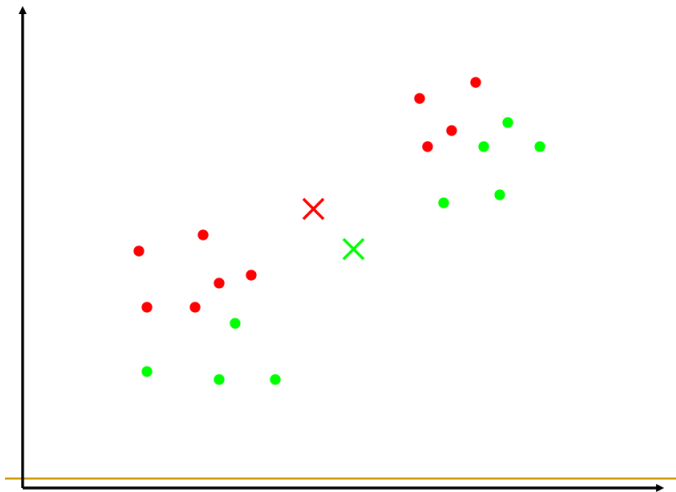


# Segmentation using K-Means Clustering (cont.)

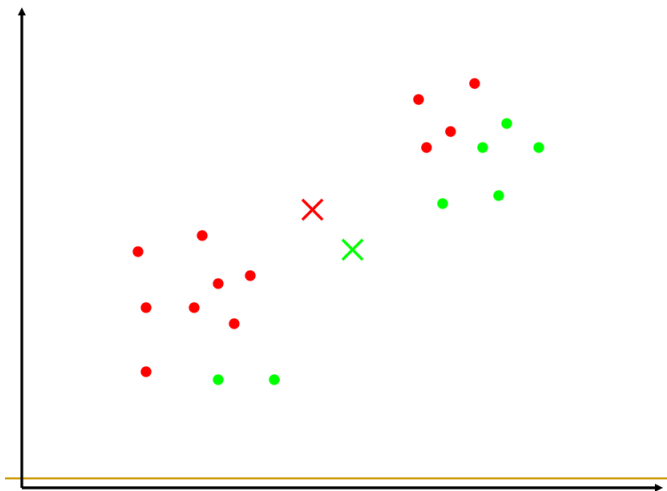




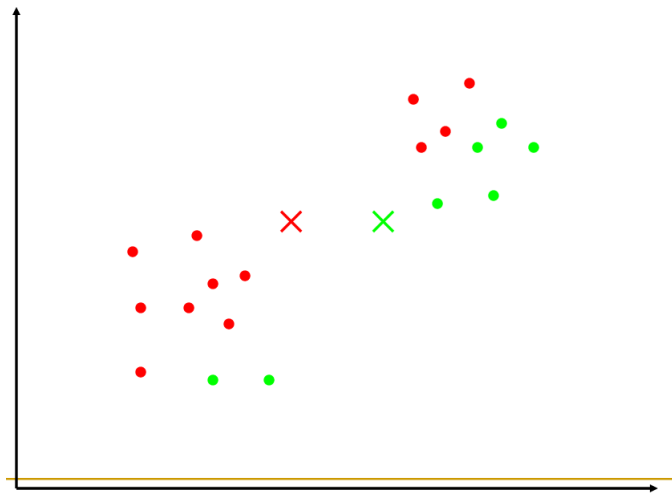
# Segmentation using K-Means Clustering (cont.)



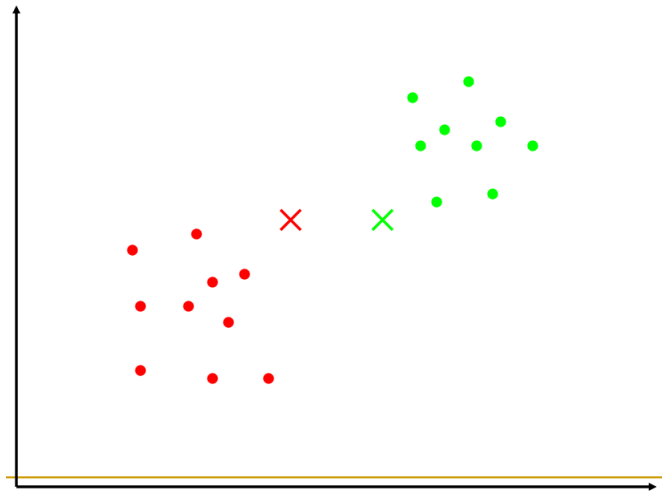
# Segmentation using K-Means Clustering (cont.)



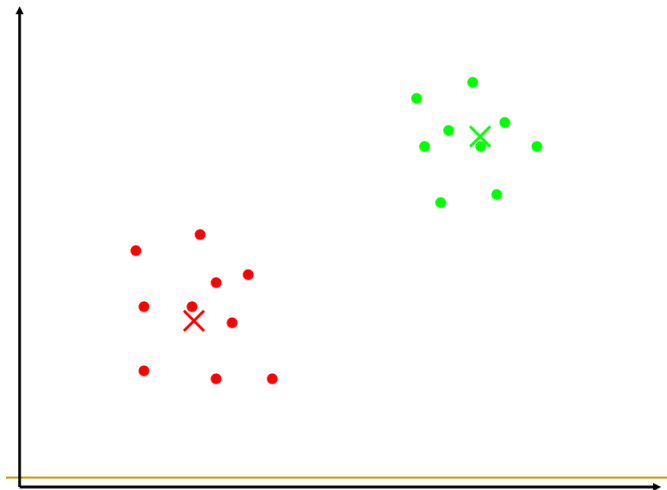
# Segmentation using K-Means Clustering (cont.)



# Segmentation using K-Means Clustering (cont.)



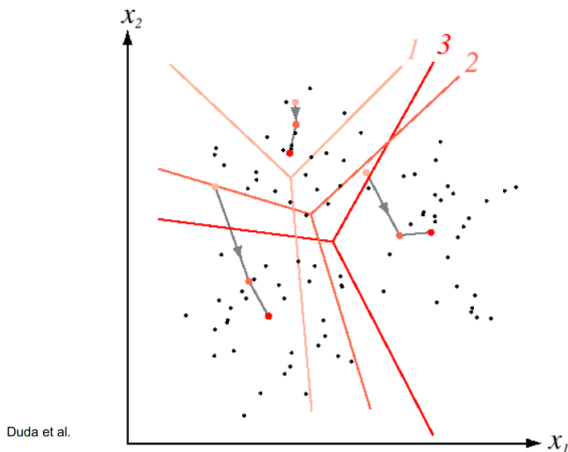
# Segmentation using K-Means Clustering (cont.)



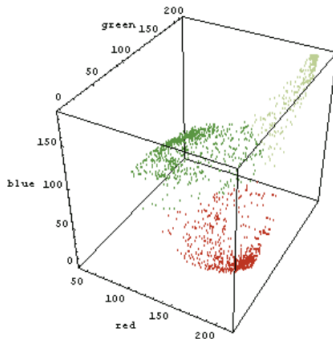
# Segmentation using K-Means Clustering (cont.)



## ► Example



► RGB vector



► K-means clustering minimizes

$$\sum_{i \in \text{clusters}} \left\{ \sum_{j \in \text{elementsofithcluster}} ||x_j - \mu_i||^2 \right\}$$

## Examples





# Segmentation using K-Means Clustering (cont.)



D. Comaniciu and P.  
Meer, *Robust Analysis  
of Feature Spaces:  
Color Image  
Segmentation*, 1997.



# Segmentation using K-Means Clustering (cont.)



Original



K=5



K=11



## K-Means Clustering: Axis Scaling

- ▶ Features of different types may have different scales.
  - For example, pixel coordinates on a 100x100 image vs. RGB color values in the range  $[0,1]$ .
- ▶ Problem: Features with larger scales dominate clustering.
- ▶ Solution: Scale the features.